

Free System Design Interview Prep Resources

Best free HLD and LLD articles, blogs, playlists, and handbooks

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Cost filter: Free only

How this shortlist was chosen

I searched across GitHub, YouTube, interview guides, engineering blogs, and public search results for resources that repeatedly appear in strong interview-prep recommendations.

Public LinkedIn reaction counts are not consistently exposed in open search results, so I ranked resources using public signals that are easier to verify: GitHub stars, playlist/video view counts, breadth of coverage, freshness, and interview relevance.

The links below are all free to access at the time of checking.

At-a-glance picks

Area	Top pick	Best for	Cost	Signal
HLD	System Design Primer	Core prep + interview questions	Free	341k stars
HLD	Gaurav Sen playlist	Concept building	Free	1M+ views on core videos
HLD	Hello Interview walkthroughs	Interview-style answers	Free	673k playlist views
LLD	awesome-low-level-design	Structured LLD topics	Free	22.9k stars
LLD	Coder Army course	OOD + UML + patterns	Free	447k intro views
LLD	Refactoring.Guru	Patterns + SOLID refresh	Free	Classic reference

Best overall free picks

Best HLD resource overall: [System Design Primer](#). About 341k GitHub stars. Still one of the strongest free resources because it bundles a study guide, interview approach advice, system design questions with solutions, object-oriented design questions with solutions, and Anki flashcards.

Best all-in-one HLD companion: [awesome-system-design-resources](#). About 35.6k GitHub stars. Organizes free material across scalability, networking, APIs, databases, architecture patterns, interview problems, engineering articles, and classic distributed-systems papers.

Best roadmap: [roadmap.sh System Design](#). Best when you want a clean sequence instead of a giant reading list. Useful for structuring HLD prep into architecture, components, interfaces, data, scalability, performance, and security.

Best LLD resource overall: [awesome-low-level-design](#). About 22.9k GitHub stars. Covers OOP fundamentals, class relationships, design principles, design patterns, UML, concurrency, and interview problems.

Best LLD repo for machine-coding style prep: [low-level-design-primer](#). About 7.6k GitHub stars. Focused on LLD questions, solutions, and scalable-system-oriented LLD preparation.

Best free courses and playlists

Best free HLD concepts course: [Gaurav Sen – System Design Playlist](#). Excellent for understanding load balancing, consistent hashing, message queues, microservices, NoSQL, and interview patterns. Public results show very strong view counts on the playlist and its core videos.

Best free HLD interview walkthrough course: [Hello Interview – System Design Walkthroughs](#). About 673,857 playlist views. One of the best free resources for learning how to structure your answer in an actual interview, not just memorize concepts.

Best free structured LLD course: [Coder Army – System Design Full Course](#). Strong for LLD fundamentals, OOD, UML, SOLID, design patterns, and real interview-flavored design exercises. The intro and foundation videos have strong public view counts.

Best free LLD plus multithreading course: [codeWithAryan – Low Level Design + MultiThreading Course](#). Very useful for companies that mix LLD with threading or concurrency questions. Covers patterns, machine-coding-style problems, and a long concurrency crash course.

Best free articles and blogs

Best fresh interview guide: [Tech Interview Handbook – System design interview guide](#). Useful for understanding the types of system design interviews: back-end or distributed design, API design, object-oriented design, and front-end system design.

Best beginner-to-intermediate written handbook: [freeCodeCamp – System Design Interview Question Handbook](#). A broad foundation across networks, storage, latency, throughput, availability, caching, proxies, load balancing, consistent hashing, databases, leader election, sockets, streaming, and pub-sub.

Best LLD / design-patterns reference: [Refactoring.Guru – Design Patterns](#). Excellent for LLD because it explains design patterns as reusable solutions to common software-design problems and organizes the classic catalog clearly.

Best deep HLD thinking resource: [Martin Fowler – Catalog of Patterns of Distributed Systems](#). Not interview-cram material, but ideal once you know the basics and want sharper language for trade-offs and distributed-systems patterns.

Best interview-structure refresher: [freeCodeCamp – Beginner's Guide to System Design](#).

Helpful if you want a simpler entry point before going deeper into handbooks, repositories, and long-form walkthroughs.

Best real-world engineering blogs for senior-level answers

Storage and database trade-offs: [Discord – How Discord Stores Trillions of Messages](#). Great for thinking about partitions, hot keys, IDs, storage evolution, and why teams migrate databases as scale changes.

Real-time messaging architecture: [Slack Engineering – Real-time Messaging](#). Very useful for websocket or gateway design, channel routing, consistent hashing, and regional broadcast patterns.

Media and file-upload systems: [Canva Engineering – From Zero to 50 Million Uploads per Day](#). Useful for object storage, metadata, pipelines, scaling uploads, and service boundary decisions.

Async event-driven design: [AWS Architecture Blog – Let's Architect! Designing event-driven architectures](#). A strong practical read for async communication, loose coupling, resilience, and event-driven trade-offs.

My strongest shortlist

HLD

Use this order: [System Design Primer](#), [Gaurav Sen](#), [Hello Interview Walkthroughs](#), then selected real-world engineering blogs from Discord, Slack, Canva, and AWS.

LLD

Use this order: [awesome-low-level-design](#), [Coder Army](#), and [Refactoring.Guru](#) for pattern and SOLID revision. Add [codeWithAryan](#) if you want extra multithreading practice.

Direct links appendix

[System Design Primer](#)

[awesome-system-design-resources](#)

[roadmap.sh System Design](#)

[awesome-low-level-design](#)

[low-level-design-primer](#)

[Gaurav Sen – System Design Playlist](#)

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[Slack Engineering – Real-time Messaging](#)
[Canva Engineering – From Zero to 50 Million Uploads per Day](#)
[AWS Architecture Blog – Let's Architect! Designing event-driven architectures](#)